



IX. Teaching And Learning Activities

9.1 Pre-assessment

Title of Pre-assessment : **BASIC Linear Algebra Concepts**

Rationale : This pretest aims to:

- Test the security of the students basic knowledge of Linear Algebra by setting questions on linear algebra that are based on their previous knowledge of secondary school mathematics.
- Find out the strngths and weaknesses of the different students in Linear Algebra so as to guide programme developers on the knowledge that the students bring with them to the course.

Questions

Linear algebra

Pretest

1. The solution to the simultaneous equations

$$x + y = 1$$

$$2x + 3y = 7$$

Is:

- A. (-5,4)
- B. (4,-5)
- C. (-4,5)
- D. (-4,-5)



2. The system

$$x + 2y = 3$$

$$3x + 6y = 9$$

Has

- A. no solution
- B. A unique solution
- C. Many solutions
- D. Zero solutions

3. The length of the vector

$$\begin{pmatrix} 5 \\ 12 \end{pmatrix}$$

Is

- A. $\begin{pmatrix} 17 \end{pmatrix}$
- B. 13
- C. 7
- D. 60

4. Which of the following has no solution

A. $\begin{pmatrix} 3 \\ 4 \\ 6 \end{pmatrix} + \begin{pmatrix} 4 \\ 0 \\ 0 \end{pmatrix}$

B. $\begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}$

C. $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

D. $\begin{bmatrix} 4 \\ 6 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \end{bmatrix}$



5. Which of the following has no solution

A. $\begin{bmatrix} 2 & 3 \\ 2 & 1 \\ 4 & 5 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 2 \\ 0 & 3 & 7 \\ 1 & 4 & 5 \end{bmatrix}$

B. $\begin{bmatrix} 2 & 3 & 1 \end{bmatrix} \times \begin{bmatrix} 7 & 1 \\ 4 & 3 \\ 2 & 1 \end{bmatrix}$

C. $\begin{bmatrix} a & x \\ c & d \end{bmatrix} \times \begin{bmatrix} x \\ z \end{bmatrix}$

D. None of the above

6. Which of the following matrices do not have a determinant

A. $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

B. $\begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 3 & 5 \\ 6 & 4 & 1 \end{bmatrix}$

D. $\begin{bmatrix} 0 & 8 & 7 \\ 7 & 1 & 9 \\ 5 & 6 & 4 \end{bmatrix}$

7. The inverse of the matrix

$$\begin{bmatrix} 2 & 5 \\ \frac{2}{5} & 1 \end{bmatrix}$$

Under multiplication is

A. $\begin{bmatrix} 1 & -5 \\ -\frac{2}{5} & 2 \end{bmatrix}$

B. $\begin{bmatrix} -2 & -5 \\ \frac{-2}{5} & -1 \end{bmatrix}$

C. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

D. Non existent



8. The matrix representing the linear transformation for a 90° anticlockwise rotation through the origin is

A. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

B. $\begin{bmatrix} -1 & 0 \\ 1 & 0 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

D. $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

9. The matrix transformation representing a reflection in the line $y=x$ is

A. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

B. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

C. $\begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix}$

D. $\begin{bmatrix} \cos 60 & \sin 60 \\ -\sin 60 & \cos 60 \end{bmatrix}$

10. The matrix that reflects an enlargement factor of k , centre $(0,0)$ is

A. $\begin{bmatrix} k & k \\ k & k \end{bmatrix}$

B. $\begin{bmatrix} 0 & k \\ k & 0 \end{bmatrix}$

C. $\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$

D. $k \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$



Answer Key

1. C
2. C
3. B
4. B
5. D
6. C
7. A
8. A
9. A
10. D

Pedagogical Comment For Learners

This pretest aims to find out whether your knowledge of basic concepts of linear algebra that you acquired during your primary and secondary school and you should sail through the test!

Should you score a score of less than 8, then you are advised to seriously revisit your secondary school materials on the topic and revise this with the help of a friend or a local secondary school teacher in your area.

Ideally, though, the content of the test is so basic and elementary that a score of less than 10 should be a cause for concern... GOOD LUCK!!!